

Linear Statistical Models W8 GA

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```
set.seed(42)
```

W8Q1a

```
"  
(a) If  $\epsilon_i \sim N(0, \sigma^2)$ , then find the distribution of  $y_i$ . Elaborate on  
your answer  
"
```

We are given the linear model $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ and ϵ_i is distributed as $N(0, \sigma^2)$.

For any given i , the values β_0 , β_1 , and x_i are treated as fixed constants. Therefore, y_i is a linear transformation of the normally distributed random variable ϵ_i .

A linear transformation of a normal random variable is also normally distributed. We can find its parameters as follows:

1. Expected Value (Mean):

$$\begin{aligned} E[y_i] &= E[\beta_0 + \beta_1 x_i + \epsilon_i] \\ E[y_i] &= E[\beta_0] + E[\beta_1 x_i] + E[\epsilon_i] \\ E[y_i] &= \beta_0 + \beta_1 x_i + 0 \\ E[y_i] &= \beta_0 + \beta_1 x_i \end{aligned}$$

2. Variance:

$$\begin{aligned} \text{Var}(y_i) &= \text{Var}(\beta_0 + \beta_1 x_i + \epsilon_i) \\ \text{Var}(y_i) &= \text{Var}(\beta_0) + \text{Var}(\beta_1 x_i) + \text{Var}(\epsilon_i) \\ \text{Var}(y_i) &= 0 + 0 + \sigma^2 \\ \text{Var}(y_i) &= \sigma^2 \end{aligned}$$

Therefore, the distribution of y_i is: y_i is distributed as

$$N(\beta_0 + \beta_1 x_i, \sigma^2).$$

```
# W8Q1b
```

```
"  
  
Using R sample one value each for  $\beta_0$ ,  $\beta_1$  and
```

one set of $\mathbf{x}_i$ as generated in code draft provided

```
"  
  
beta_0 <- runif(1, 5, 12)  
  
beta_1 <- runif(1, -10, 10)  
  
N <- 1000  
  
x <- runif(N, -5, 5)  
  
print(paste("True beta_0:", beta_0))  
  
print(paste("True beta_1:", beta_1))  
  
print(paste("N:", N))  
  
print(paste("x:", x))
```

```
[1] "\nUsing R sample one value each for  $\beta_0$ ,  $\beta_1$  and\none set of  $\mathbf{x}_i$  as generated in code dra  
ft provided\n"  
[1] "True beta_0: 9.07455891673453"  
[1] "True beta_1: -6.28388644661754"  
[1] "N: 1000"  
[1] "x: 2.41125292843208"      "x: 3.55072640581056"      "x: 3.08614781359211"  
[4] "x: -3.57331012608483"     "x: 1.74663689685985"     "x: 0.198947822209448"  
[7] "x: -0.106566166505218"    "x: -2.1473133424297"     "x: -2.67759225796908"  
[10] "x: -3.08734270045534"     "x: 0.444630859419703"    "x: 4.05140741262585"  
[13] "x: 3.57514981180429"     "x: -1.31294751772657"    "x: 0.00101154902949929"  
[16] "x: -4.24660252872854"     "x: -4.74941897206008"    "x: -4.59610713878646"  
[19] "x: -0.577915257308632"    "x: -2.18895183410496"    "x: 0.815765387378633"  
[22] "x: -2.66908987192437"     "x: -3.91866930527613"    "x: -1.0692752734758"  
[25] "x: 4.4857851555571"       "x: 3.30507626291364"     "x: -2.41786625469103"  
[28] "x: -3.89942756853998"     "x: -4.62563762441278"    "x: -3.04368196288124"  
[31] "x: 4.03939390322194"     "x: 1.79385776631534"     "x: -3.04270405322313"  
[34] "x: -0.655437882523984"    "x: -2.07549002021551"    "x: 4.17796613881364"  
[37] "x: 3.95734818652272"     "x: -2.46800377266482"    "x: -1.9235702441074"
```

```
# W8Q1c  
"  
Using R, generate the dataset, i.e.  $\mathbf{y}_i$   
, and store it in a data frame  
"
```

```
error <- rnorm(N, 0, 1)
y <- beta_0 + beta_1 * x + error
dataset <- data.frame(x, y)

View(dataset)
```

dataset		
(row)	x	y
1	2.4113	-5.4119
2	3.5507	-14.2374
3	3.0861	-8.3375
4	-3.5733	33.321
5	1.7466	-2.7933
6	0.1989	6.9706
7	-0.1066	10.6159
8	-2.1473	23.4599
9	-2.6776	25.7672
10	-3.0873	28.5258
11	0.4446	7.8439

W8Q1d

```

"
Perform 200 iterations of the below and store the values of
 $\beta_1$ ,  $R_1^2$ ,  $R_0^2$  and  $R_1^2 - R_0^2$  computed in a data frame called
estimates.
"
"
(i) Sample  $n = 100$  members for the data. Find the least square estimates
of  $\beta_0$ 
and  $\beta_1$  for this sampled data.
"

# W8Q1d(i)

n_iterations <- 200
n <- 100

# Initialize a data frame to store results
estimates <- data.frame(
  beta_1_hat = numeric(n_iterations),
  R_0_square = numeric(n_iterations),
  R_1_square = numeric(n_iterations),
  Reg_SS = numeric(n_iterations) #  $R_1^2 - R_0^2$ 
)

for (i in 1:n_iterations) {
  # i. Sample  $n=100$  members
  sampled_data <- dataset[sample(1:N, n), ]

  x_sample <- sampled_data$x
  y_sample <- sampled_data$y

  # Calculate means
  x_bar <- mean(x_sample)
  y_bar <- mean(y_sample)

  # Calculate LSEs
  beta_1_hat <- sum((y_sample - y_bar) * (x_sample - x_bar)) /
sum((x_sample - x_bar)^2)

```

```

beta_0_hat <- y_bar - beta_1_hat * x_bar

# ii. Compute Sum of Squares
# R_1^2 (Total Sum of Squares, TSS)
R_1_square <- sum((y_sample - y_bar)^2)

# R_0^2 (Residual Sum of Squares, RSS)
# Using the formula from the draft code
R_0_square <- sum((y_sample - y_bar)^2) - (sum((y_sample - y_bar) *
(x_sample - x_bar)))^2 / sum((x_sample - x_bar)^2)

# C. R_1^2 - R_0^2 (Regression Sum of Squares, RegSS)
Reg_SS <- R_1_square - R_0_square

# iii. Estimate sigma^2
sigma_2_hat <- R_0_square / (n - 2)

# Store values
estimates$beta_1_hat[i] <- beta_1_hat
estimates$R_0_square[i] <- R_0_square
estimates$R_1_square[i] <- R_1_square
estimates$Reg_SS[i] <- Reg_SS
}

print("Q1 Iterations complete. Displaying plots...")

```

```

R_0_square <- sum((y_sample - y_bar)^2) - (sum((y_sample - y_bar) * (x_sample - x_bar)))^2 / sum((x_sample - x_bar)^2)

# C.  $R_1^2 - R_0^2$  (Regression Sum of Squares, RegSS)
Reg_SS <- R_1_square - R_0_square

# iii. Estimate  $\sigma^2$ 
sigma_2_hat <- R_0_square / (n - 2)

# Store values
estimates$beta_1_hat[i] <- beta_1_hat
estimates$R_0_square[i] <- R_0_square
estimates$R_1_square[i] <- R_1_square
estimates$Reg_SS[i] <- Reg_SS
}

print("Q1 Iterations complete. Displaying plots...")
1] "Q1 Iterations complete. Displaying plots..."

```

1(d)-iii. Estimate σ^2 and find the Distribution of β_1_hat

- Estimate of σ^2 : The unbiased estimator for the variance σ^2 in a simple linear regression model is the Residual Sum of Squares (R_0^2) divided by its degrees of freedom, which are $(n-2)$.
 $\sigma^2_hat = R_0^2 / (n-2)$
- Distribution of β_1_hat : The least-squares estimator β_1_hat is a linear combination of the y_i variables. Since each y_i is normally distributed (as shown in part 1a), β_1_hat itself is also normally distributed.
 1. Expected Value: $E[\beta_1_hat] = \beta_1$ (It is an unbiased estimator).
 2. Variance: $Var(\beta_1_hat) = \sigma^2 / \sum (x_i - \bar{x})^2$
- Therefore, the distribution of the estimator β_1_hat is: β_1_hat is distributed as $N(\beta_1, \sigma^2 / \sum (x_i - \bar{x})^2)$

```

# W8Q1e and # W8Q1f
# 1(e) & (f) Plot histograms and fit densities
# Set up 2x2 plotting area
par(mfrow = c(2, 2))

# 1. beta_1_hat
hist(estimates$beta_1_hat,
     prob = TRUE,
     main = expression(paste("Histogram of ", hat(beta)[1])),
     xlab = expression(hat(beta)[1]), breaks = 20, col = "lightblue"
)

# Overlay Normal density (using estimated parameters from the
simulation)
curve(dnorm(x, mean = mean(estimates$beta_1_hat), sd =
sd(estimates$beta_1_hat)),

```

```

    add = TRUE, col = "red", lwd = 2
)
# Add vertical line for the true beta_1
abline(v = beta_1, col = "blue", lwd = 2, lty = 2)
legend("topright", legend = c("Fitted Density", "True beta_1"), col =
c("red", "blue"), lty = c(1, 2))

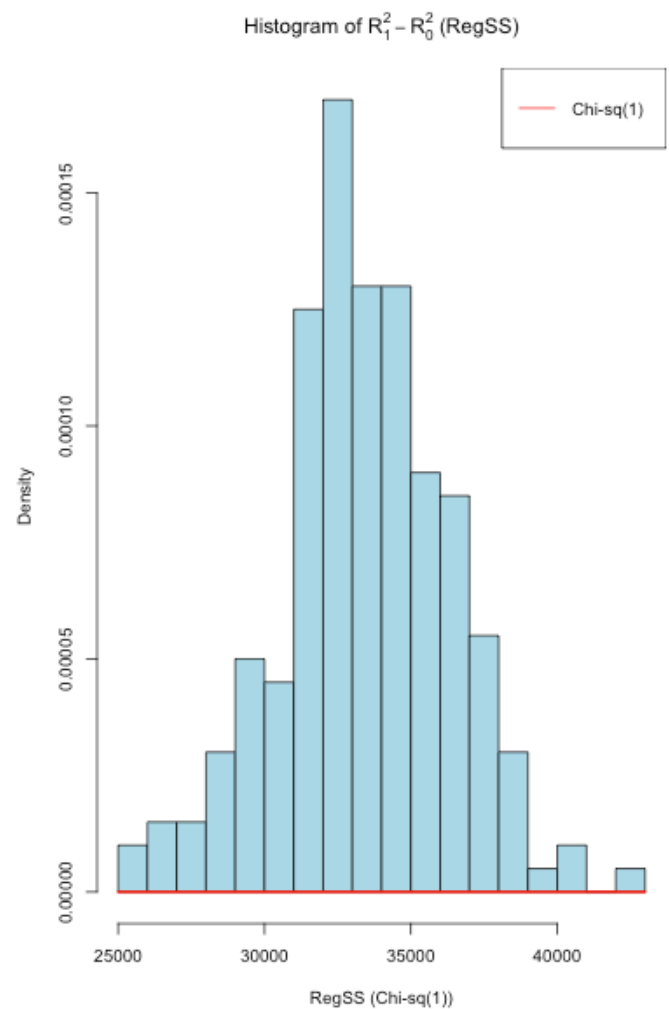
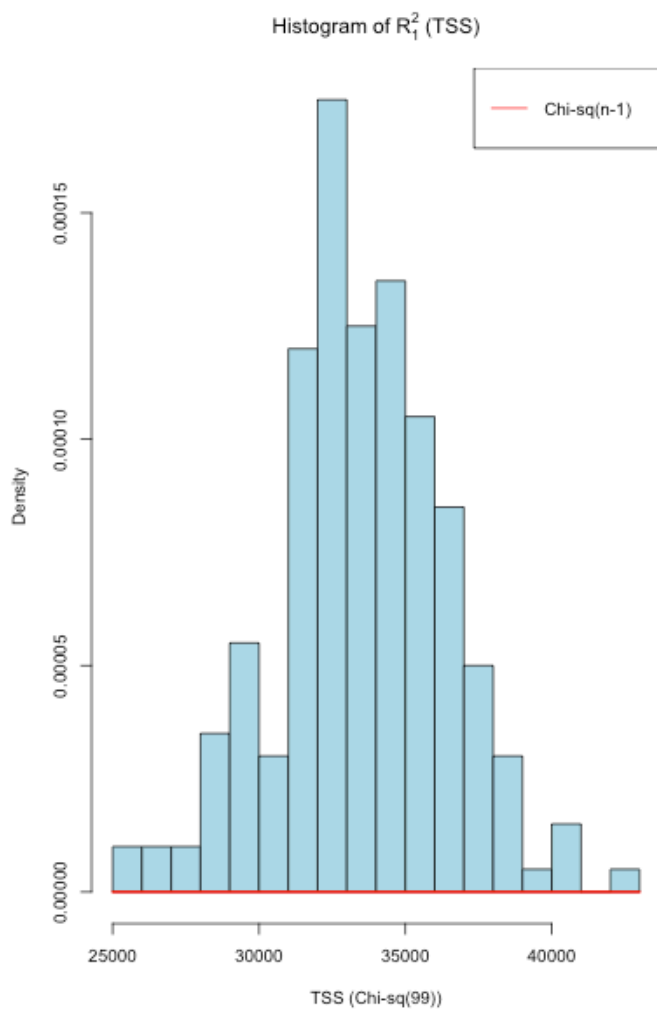
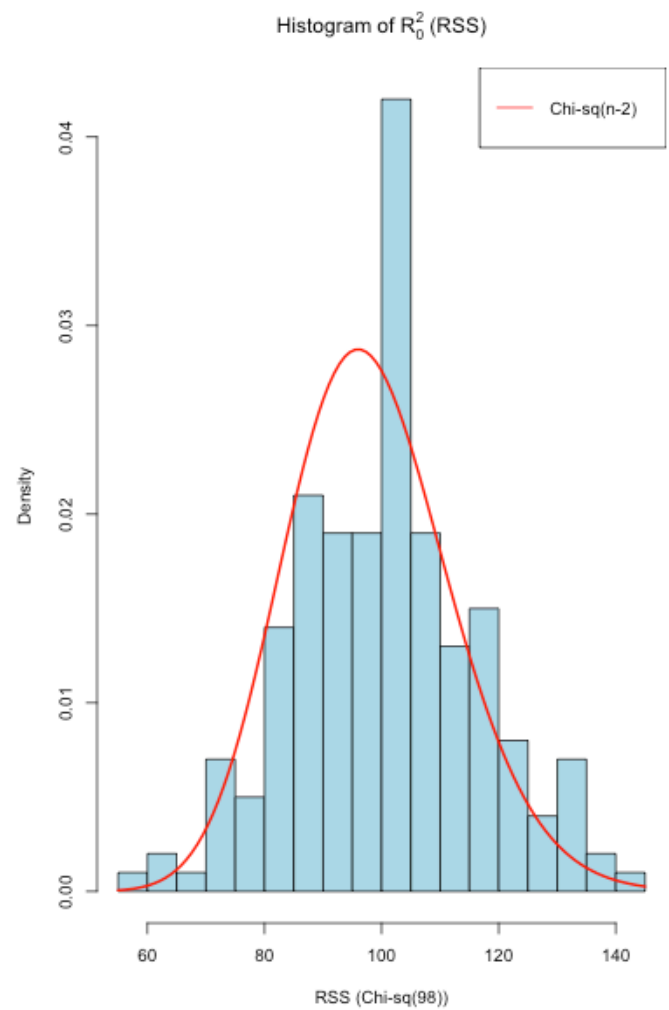
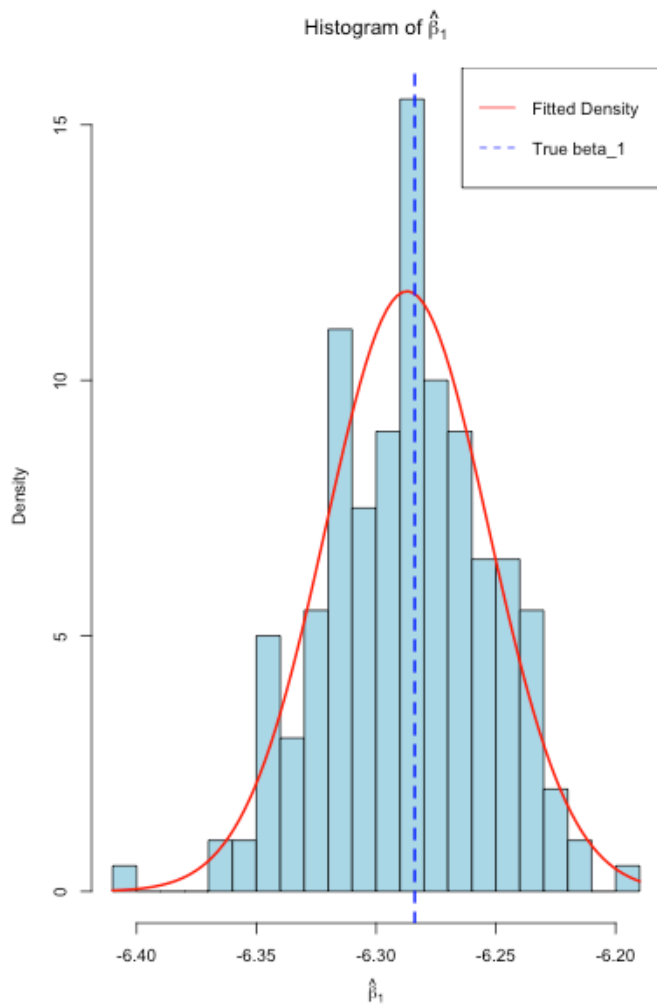
# 2. R_0^2 (RSS)
hist(estimates$R_0_square,
     prob = TRUE,
     main = expression(paste("Histogram of ", R[0]^2, " (RSS)")),
     xlab = "RSS (Chi-sq(98))", breaks = 20, col = "lightblue"
)
# Overlay Chi-square(n-2) density (df = 98)
curve(dchisq(x, df = n - 2), add = TRUE, col = "red", lwd = 2)
legend("topright", legend = c("Chi-sq(n-2)"), col = "red", lty = 1)

# 3. R_1^2 (TSS)
hist(estimates$R_1_square,
     prob = TRUE,
     main = expression(paste("Histogram of ", R[1]^2, " (TSS)")),
     xlab = "TSS (Chi-sq(99))", breaks = 20, col = "lightblue"
)
# Overlay Chi-square(n-1) density (df = 99)
curve(dchisq(x, df = n - 1), add = TRUE, col = "red", lwd = 2)
legend("topright", legend = c("Chi-sq(n-1)"), col = "red", lty = 1)

# 4. R_1^2 - R_0^2 (RegSS)
hist(estimates$Reg_SS,
     prob = TRUE,
     main = expression(paste("Histogram of ", R[1]^2 - R[0]^2, "
(RegSS)")),
     xlab = "RegSS (Chi-sq(1))", breaks = 20, col = "lightblue"
)
# Overlay Chi-square(1) density (df = 1)
curve(dchisq(x, df = 1), add = TRUE, col = "red", lwd = 2)
legend("topright", legend = c("Chi-sq(1)"), col = "red", lty = 1)

```

```
# Reset plotting layout  
par(mfrow = c(1, 1))
```

1(f)

1. β_1 Histogram: The histogram for β_1 aligns very well with the overlaid normal distribution curve. The mean of the simulated β_1 values is very close to the true β_1 parameter, visually confirming that β_1 is an unbiased, normally distributed estimator.
2. R_0^2 (RSS) Histogram: The histogram for R_0^2 (Residual Sum of Squares) closely follows the Chi-square distribution with $(n-2) = 98$ degrees of freedom. This is expected because (R_0^2 / σ^2) is distributed as $\text{Chi-square}(n-2)$, and in this simulation, $\sigma^2 = 1$.
3. R_1^2 (TSS) Histogram: The histogram for R_1^2 (Total Sum of Squares) does NOT fit the overlaid Chi-square distribution with $(n-1) = 99$ degrees of freedom. This distribution is only correct under the null hypothesis that $\beta_1 = 0$. Since we sampled a non-zero β_1 , the histogram is shifted significantly to the right of the central Chi-square curve.
4. $R_1^2 - R_0^2$ (RegSS) Histogram: The histogram for the Regression Sum of Squares ($R_1^2 - R_0^2$) does NOT fit the Chi-square distribution with 1 degree of freedom. This distribution is also only correct under the null hypothesis $\beta_1 = 0$. Since β_1 is not zero, the histogram is shifted far to the right.

```
#####
# QUESTION 2
#####

"
Consider the Linear Model:


$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \varepsilon_i; 1 \leq i \leq 1000$$


"

# W8Q2a

"
If  $\varepsilon_i \sim N(0, \sigma^2)$ , then find the distribution of  $y_i$ .
Elaborate on your answer.
"

# Done on Report
```

2(a) Distribution of y_i

You are given the model $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \varepsilon_i$ and ε_i is distributed as $N(0, \sigma^2)$.

Similar to Question 1, y_i is a linear transformation of the normally distributed variable ε_i .

1. Expected Value (Mean): $E[y_i] = E[\beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \varepsilon_i]$ $E[y_i] = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + E[\varepsilon_i]$ $E[y_i] = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i}$
2. Variance: $\text{Var}(y_i) = \text{Var}(\beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \varepsilon_i)$ $\text{Var}(y_i) = \text{Var}(\varepsilon_i) = \sigma^2$

Therefore, the distribution of y_i is: y_i is distributed as $N(\beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i}, \sigma^2)$.

```
# W8Q2b

"

Using R sample one value each for  $\beta_0, \beta_1, \beta_2$ 
and one set of  $x_{1i}$  and  $x_{2i}$  as generated in code draft provided.
```

```

"
beta_0_2 <- runif(1, 5, 12)
beta_1_2 <- runif(1, -10, 10)
beta_2_2 <- runif(1, 2, 7)
N_2 <- 1000
x_1 <- runif(N_2, -5, 5)
x_2 <- runif(N_2, 0, 10)

print(paste("Q2 True beta_0:", beta_0_2))
print(paste("Q2 True beta_1:", beta_1_2))
print(paste("Q2 True beta_2:", beta_2_2))

```

```

    print(paste("Q2 True beta_0:", beta_0_2))
    print(paste("Q2 True beta_1:", beta_1_2))
    print(paste("Q2 True beta_2:", beta_2_2))
[1] "\nUsing R sample one value each for  $\beta_0$ ,  $\beta_1$ ,  $\beta_2$ \nand one set of  $x_{1i}$  and  $x_{2i}$  as genera
ted in code draft provided.\n"
[1] "Q2 True beta_0: 9.76502312370576"
[1] "Q2 True beta_1: 7.01970267109573"
[1] "Q2 True beta_2: 3.58755073440261"

```

```

# W8Q2c
error_2 <- rnorm(N_2, 0, 1) # sigma^2 is 1
y_2 <- beta_0_2 + beta_1_2 * x_1 + beta_2_2 * x_2 + error_2
dataset_2 <- data.frame(x_1, x_2, y = y_2) # Renaming y_2 to y for model
fitting
View(dataset_2)

```

```

r$> # W8Q2c
error_2 <- rnorm(N_2, 0, 1) # sigma^2 is 1
y_2 <- beta_0_2 + beta_1_2 * x_1 + beta_2_2 * x_2 + error_2
dataset_2 <- data.frame(x_1, x_2, y = y_2) # Renaming y_2 to y for model fitting
View(dataset_2)

```

 *plot.png* dataset_2 ×

(row)	x_1	x_2	y
1	0.9386	7.4415	42.8184
2	4.0864	5.3702	57.61
3	2.4447	4.9314	43.529
4	-4.9685	0.89	-20.0683
5	0.0516	3.4599	21.9696
6	4.9815	8.3646	74.5087
7	-1.8702	8.5795	26.1967
8	0.9786	1.4587	22.4402
9	-0.077	8.928	39.479
10	0.2347	3.1719	23.8463
11	4.1656	0.1621	40.8204

W8Q2d

n_iterations_2 <- 200

n_2 <- 100

Initialize data frame

estimates2 <- data.frame(

R_0_square = numeric(n_iterations_2),

R_1_square = numeric(n_iterations_2),

Reg_SS = numeric(n_iterations_2) # $R_1^2 - R_0^2$

)

```

for (i in 1:n_iterations_2) {
  # i. Sample n=100 members
  sampled_data_2 <- dataset_2[sample(1:N_2, n_2), ]

  # Find LSEs (using R's built-in lm() function for simplicity and
stability)
  # This is equivalent to the matrix algebra in the draft
  model <- lm(y ~ x_1 + x_2, data = sampled_data_2)

  # Calculate predicted values
  y_hat <- model$fitted.values
  y_sample <- sampled_data_2$y

  # ii. Compute Sum of Squares
  y_bar <- mean(y_sample)

  # A.  $R_0^2$  (Residual Sum of Squares, RSS)
  R_0_square <- sum((y_sample - y_hat)^2) # Same as
sum(model$residuals^2)

  # B.  $R_1^2$  (Total Sum of Squares, TSS)
  R_1_square <- sum((y_sample - y_bar)^2)

  # C.  $R_1^2 - R_0^2$  (Regression Sum of Squares, RegSS)
  Reg_SS <- R_1_square - R_0_square

  # Store values
  estimates2$R_0_square[i] <- R_0_square
  estimates2$R_1_square[i] <- R_1_square
  estimates2$Reg_SS[i] <- Reg_SS
}

```

```

# A.  $R_0^2$  (Residual Sum of Squares, RSS)
R_0_square <- sum((y_sample - y_hat)^2) # Same as sum(model$residuals^2)

# B.  $R_1^2$  (Total Sum of Squares, TSS)
R_1_square <- sum((y_sample - y_bar)^2)

# C.  $R_1^2 - R_0^2$  (Regression Sum of Squares, RegSS)
Reg_SS <- R_1_square - R_0_square

# Store values
estimates2$R_0_square[i] <- R_0_square
estimates2$R_1_square[i] <- R_1_square
estimates2$Reg_SS[i] <- Reg_SS
}

```

```

# W8Q2e

# Set up 1x3 plotting area
par(mfrow = c(1, 3))

# 1.  $R_0^2$  (RSS)
hist(estimates2$R_0_square,
     prob = TRUE,
     main = expression(paste("Histogram of ", R[0]^2, " (RSS)")),
     xlab = "RSS (Chi-sq(97))", breaks = 20, col = "lightgreen"
)

# Overlay Chi-square(n-3) density (df = 97)
curve(dchisq(x, df = n_2 - 3), add = TRUE, col = "blue", lwd = 2)
legend("topright", legend = c("Chi-sq(n-3)"), col = "blue", lty = 1)

# 2.  $R_1^2$  (TSS)
hist(estimates2$R_1_square,
     prob = TRUE,
     main = expression(paste("Histogram of ", R[1]^2, " (TSS)")),
     xlab = "TSS (Chi-sq(99))", breaks = 20, col = "lightgreen"
)

# Overlay Chi-square(n-1) density (df = 99)
curve(dchisq(x, df = n_2 - 1), add = TRUE, col = "blue", lwd = 2)
legend("topright", legend = c("Chi-sq(n-1)"), col = "blue", lty = 1)

```

```
# 3.  $R_1^2 - R_0^2$  (RegSS)
hist(estimates2$Reg_SS,
     prob = TRUE,
     main = expression(paste("Histogram of ",  $R[1]^2 - R[0]^2$ , "
(RegSS)")),
     xlab = "RegSS (Chi-sq(2))", breaks = 20, col = "lightgreen"
)
# Overlay Chi-square(2) density (df = 2)
curve(dchisq(x, df = 2), add = TRUE, col = "blue", lwd = 2)
legend("topright", legend = c("Chi-sq(2)"), col = "blue", lty = 1)

# Reset plotting layout
par(mfrow = c(1, 1))
```

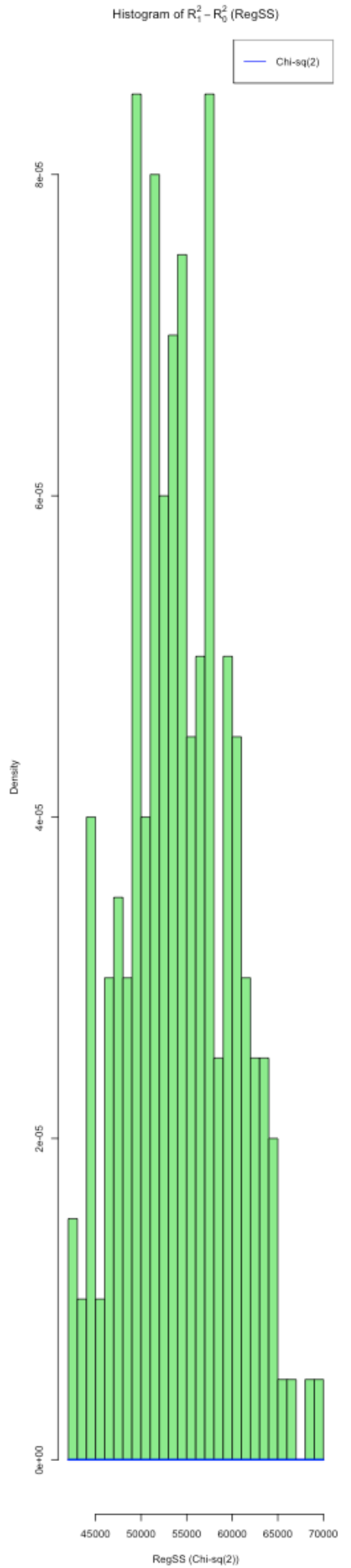
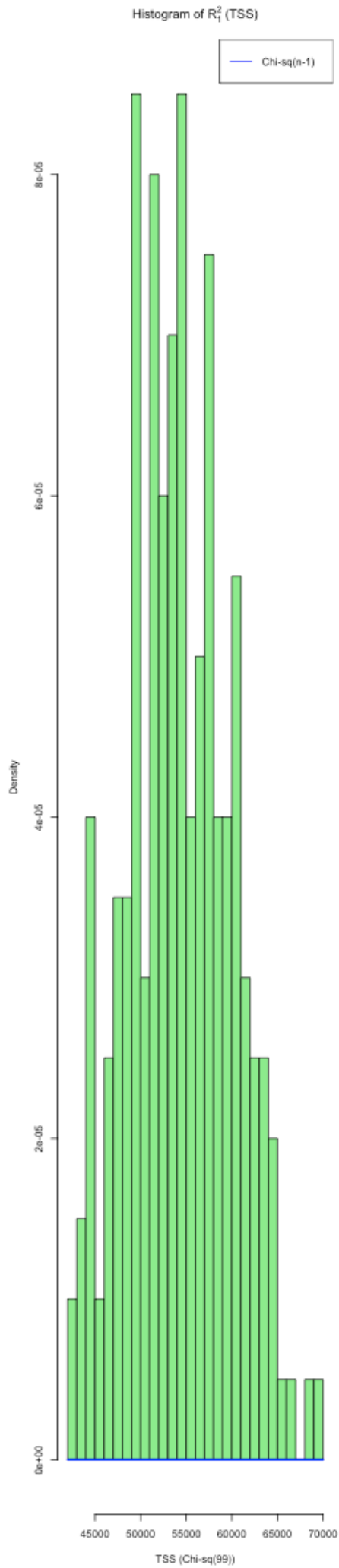
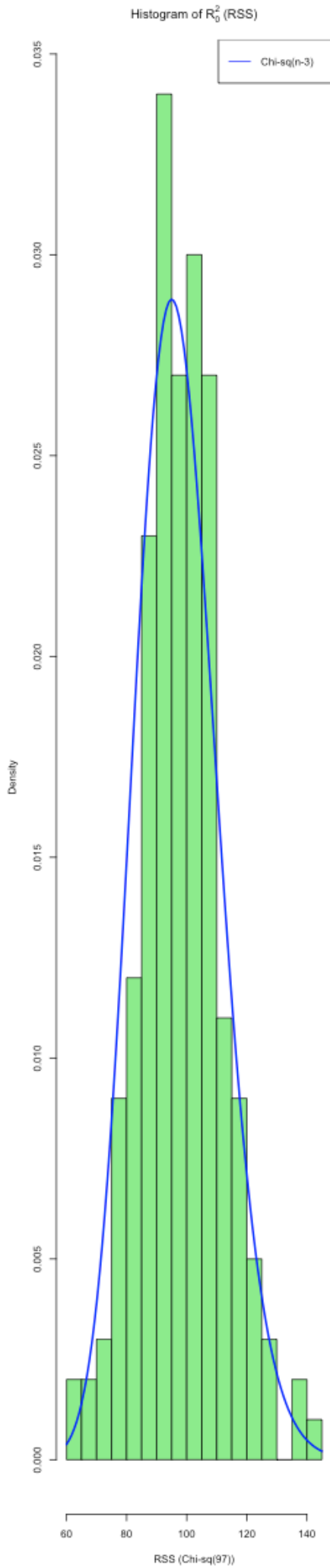
PROBLEMS 42 OUTPUT TERMINAL PORTS DEBUG CONSOLE R Interactive

```

    prob = TRUE,
    main = expression(paste("Histogram of ",  $R[1]^2 - R[0]^2$ , " (RegSS)")),
    xlab = "RegSS (Chi-sq(2))", breaks = 20, col = "lightgreen"
  )
  # Overlay Chi-square(2) density (df = 2)
  curve(dchisq(x, df = 2), add = TRUE, col = "blue", lwd = 2)
  legend("topright", legend = c("Chi-sq(2)"), col = "blue", lty = 1)

  # Reset plotting layout
  par(mfrow = c(1, 1))

r$> 
```

W8Q2f

1. R_0^2 (RSS) Histogram: The histogram for R_0^2 aligns very well with the Chi-square distribution with $(n-3) = 97$ degrees of freedom. The degrees of freedom are $n - p - 1$, where $n=100$ and $p=2$. This fit is expected because $\sigma^2=1$ in our simulation.
2. R_1^2 (TSS) Histogram: The histogram for R_1^2 does NOT fit the Chi-square distribution with $(n-1) = 99$ degrees of freedom. This distribution only holds under the null hypothesis H_0 : $\beta_1 = \beta_2 = 0$. Since our sampled parameters are non-zero, the histogram is shifted far to the right.
3. $R_1^2 - R_0^2$ (RegSS) Histogram: The histogram for $R_1^2 - R_0^2$ does NOT fit the Chi-square distribution with 2 degrees of freedom. This distribution only holds under the null hypothesis H_0 : $\beta_1 = \beta_2 = 0$. Our non-zero parameters cause the histogram to be shifted far to the right.